

Discharging H-diag III: the off-diagonal constant certificate – the

TECT verification-first repository · claim B2-PROPA-HLAYER

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1. Purpose and scope

The reduction note (hdiag-offdiag-additive-energy) showed the off-diagonal Bogoliubov weight equals the additive energy $\sum_Q p_2^2 = E_+$ and reduced the class-wide stability to $R \lesssim u_{\text{eff}}^2 B_{\text{max}} r_R^{-1} I(1 + T')$, verifying $R < 1$ for BCC (worst 0.916). The operator's review directed a constant certificate with four goals: pin B_{max} ; a class-wide diagonal lower bound; bound the $p_4 / p_2 p_4$ terms; and assemble the lattice-class ratio. This note executes all four and reports the honest verdict at the operating intensity.

2. The four pinned constants

(1) B_{max} . The bubble $B(|Q|) = \int_q G_d(q)G_d(q+Q) = J(|Q|, \hat{r})$ is monotone decreasing in $|Q|$, so $B_{\text{max}} = B(|Q|_{\text{min}})$. For the BCC Bragg set the smallest nonzero transfer is $|Q|_{\text{min}} = q_0 = 0.680$, giving

$$B(0) = 0.299, \quad B_{\text{max}} = B(q_0) = 0.218 \leq B(0).$$

(2) **Diagonal floor.** Math428 (§3bis) gives the small- A leading term $\Delta F_{\text{diag}}(A) = c_{\text{diag}} A^2 + O(A^4)$ with $c_{\text{diag}} = n r_R = \frac{N}{2} r_R = 1.827 (> 0)$; at finite A the diagonal term only grows, so $\Delta F_{\text{diag}} \geq c_{\text{diag}} A^2$ is a conservative class-wide lower bound.

(3) p_4 / **cross terms.** $W(Q)^2 = 9u_{\text{eff}}^2 A^4 p_2^2 + 30u_{\text{eff}} v A^6 p_2 p_4 + 25v^2 A^8 p_4^2$. Relative to the leading $A^4 p_2^2$ term, the cross and quartic carry extra $A^2 \sim I/N$ factors. Evaluated at the operating amplitude,

$$\left. \frac{30u_{\text{eff}} v A^2 \sum_Q p_2 p_4}{9u_{\text{eff}}^2 \sum_Q p_2^2} \right|_{A^2=I/N} = 0.044 \ll 1,$$

so the leading $p_2^2 = E_+$ term controls the bound at the operating intensity.

(4) **The assembled ratio.** With $A^2 = I/N$ (condensate intensity I over N modes), $E_+ \leq (1 + T'')N^2$, and the worst case $\rho = 1$,

$$R_{\text{lead}}(I) \leq \frac{\frac{1}{4} \cdot 9 u_{\text{eff}}^2 (I/N)^2 B_{\text{max}} E_+}{c_{\text{diag}}(I/N)} = \frac{9}{4} \frac{u_{\text{eff}}^2 B_{\text{max}} E_+}{c_{\text{diag}}} \frac{I}{N}.$$

3. The verdict at the operating intensity

$$R_{\text{lead}}(I) \leq \begin{cases} 0.035, & I = 4 \times 10^{-4} \\ 0.087, & I = 10^{-3} \\ 0.174, & I = 2 \times 10^{-3} \text{ (operating endpoint)} \end{cases}$$

with $A_{\text{op}} = \sqrt{I/N} = 0.013$ at $I = 2 \times 10^{-3}$.

The conservative leading class-wide ratio is ≤ 0.174 at the operating endpoint intensity – a margin of $\times 5.7$ below unity. The thinness reported for BCC ($R = 0.916$ at $A = 0.08$) lies at an amplitude $\sim 6\times$ the operating A_{op} (intensity $\sim 40\times$ larger), OUTSIDE the operating regime: because the off-diagonal share is $O(A^4)$ against the diagonal $O(A^2)$, $R = O(A^2) \rightarrow 0$ as $A \rightarrow 0$, and the operating amplitude sits at the comfortable small- A end.

4. Honest scope (operator point 4): condensate direction, not worst direction

R is the CONDENSATE-DIRECTION scalar ratio $|\Delta F_{\text{od}}[\delta G_{\text{cond}}]|/\Delta F_{\text{diag}}[\delta G_{\text{cond}}]$ for the physical Bloch perturbation δG_{cond} generated by the condensate coupling $W(Q)$. The additive-energy sum $\sum_Q W(Q)^2 B(|Q|)$ is the Cauchy–Schwarz / worst-transfer ENVELOPE over the condensate-compatible off-diagonal channels, so it upper-bounds this ratio. It is NOT the full unrestricted worst-direction operator norm $\|E^{-1/2} \text{Hess } \Phi_{\text{od}} E^{-1/2}\|_{\text{op}}$ over arbitrary δG , which requires the complete Hessian (including the entropy cross-block structure) and remains the residual. The safe statement is

$$\frac{|\Delta F_{\text{od}}[\delta G_{\text{cond}}]|}{\Delta F_{\text{diag}}[\delta G_{\text{cond}}]} \leq R_{\text{lead}}(I) \leq 0.174 \text{ (operating } I),$$

i.e. no condensate-induced Bloch competitor lowers F below the isotropic value at the operating intensity, for the lattice class, at leading order.

5. Status and next step

T4 STRONG EVIDENCE. Pinned: $B_{\text{max}} = 0.218$, $c_{\text{diag}} = 1.827$. Shown: p_4/cross suppressed (0.044). Certified: the condensate-direction class-wide ratio $R_{\text{lead}} \leq 0.174$ ($\times 5.7$) at the operating endpoint intensity. **OPEN (residuals):** (i) the full unrestricted worst-direction operator norm (the complete Hessian); (ii) a rigorous – not leading-order – p_4/cross bound via the higher additive energies $\sum p_4^2$, $\sum |p_2 p_4|$; (iii) off-anchor $u_{\text{eff}} > 0$ (ROBUSTNESS-MU2); (iv) the RES-5 beyond-second-cumulant extension. No tier flip: B2-PROPA-HLAYER T6, B1-RH-ENUM T6 on {H-LAYER}. H-diag/RES-1 is now a STRONG discharge candidate for the lattice class at the operating intensity, pending the worst-direction norm.

6. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “Leading order only – the p_4 term is not rigorously bounded.” VALID: the cross term is shown suppressed numerically (0.044) at the operating A , but a class-wide THEOREM needs $\sum_Q p_4^2$, $\sum_Q |p_2 p_4|$ bounded by the higher additive energies (the same R-026 divisor machinery extends, but is not executed here). Leading-order certificate, stated as such.

(β) “*This is the condensate-direction ratio, not the operator norm.*” VALID (operator point 4), and §4 says so explicitly: the certificate bounds the physical Bloch competitors; the full worst-direction Hessian norm is the named residual. No claim of $\|\cdot\|_{\text{op}} < 1$ in the unrestricted sense.

(γ) “ *B_{max} and c_{diag} are anchor/numerical.*” VALID-with-mitigation: $B_{\text{max}} = 0.218$ is a converged bubble quadrature (the $\rho = 1$ worst case, no resummation credit), $c_{\text{diag}} = \frac{N}{2}r_R$ is analytic; both at the anchor μ^2 , the ROBUSTNESS-MU2 axis (closed $[\times 0.5, \times 2]$, where $u_{\text{eff}} > 0$).

Quantitative sanity check (mandatory): *magnitude* – $R_{\text{lead}}(2e-3) = 0.174$, $\times 5.7$ below unity; *limit* – $R = O(A^2)$: $0.035/0.087/0.174$ scale as $I = 4/10/20 \times 10^{-4}$ (linear in I , since $A^2 = I/N$); *ordering* – the operating $A_{\text{op}} = 0.013$ is $6\times$ below the thin $A = 0.08$; *consistency* – $B_{\text{max}} = 0.218 < B(0) = 0.299$ (bubble decreasing in $|Q|$); *sign* – $\Delta F_{\text{od}} < 0$ dominated by $\Delta F_{\text{diag}} > 0$ with margin.

7. Result footer

Result ID:	B2-PROPA-HLAYER / H-diag discharge III (off-diagonal constant certificate)
Precise statement:	Constants pinned: $B_{\text{max}}=B(q_0)=0.218 \leq B(0)=0.299$; $c_{\text{diag}}=(N/2)r_R=1.827$; p_4 cross/leading=0.044 (suppressed by $A^2 \sim I/N$). Assembled conservative leading class-wide ratio $R_{\text{lead}}(I)=(9/4)u_{\text{eff}}^2 B_{\text{max}} E_+ (I/N)/c_{\text{diag}} = 0.035/0.087/0.174$ at $I=4e-4/1e-3/2e-3$, so $R_{\text{lead}} \leq 0.174$ ($\times 5.7$) at the operating endpoint. This bounds the CONDENSATE-DIRECTION ratio (additive-energy envelope), NOT the unrestricted worst-direction operator norm.
Scope:	Second-cumulant, anchor μ^2 , lattice class, leading- p_2 order, condensate-direction; operating intensity $A^2 \sim I/N$ ($A_{\text{op}}=0.013$).
Dependencies:	hdiag-offdiag-additive-energy (sum $p_2^2=E_+$, $R < 1$ BCC); Math428 (W,B, c_{diag}); sectorb_common.J_of_t (bubble); Math424-AddA (r_R , M_R , u_{eff}); R-025 ($E_+ \leq (1+T')N^2$).
Evidence grade:	ANALYTIC (bound) + EXECUTED (hdiag_offdiag_constant_certificate.py 5/5).
Reproduction command:	python codes/vacuum/hdiag_offdiag_constant_certificate.py
Expected output:	$B_{\text{max}}=0.218$; $c_{\text{diag}}=1.827$; $R_{\text{lead}}(2e-3)=0.174$; cross/lead=0.044; 5/5 PASS.
Falsification gate:	$R_{\text{lead}} > 1$ at the operating intensity; $B_{\text{max}} > B(0)$; a condensate Bloch competitor with $F < F[G^*]$ at A_{op} ; or the p_4 terms not suppressed at operating A .
Tier before / after:	No tier flip. B2 T6, B1 T6 on {H-LAYER} unchanged. H-diag/RES-1: STRONG discharge candidate for the lattice class at operating intensity (condensate-direction).
No-overclaim statement:	This does NOT discharge H-diag/H-LAYER and does NOT promote B1. It certifies the condensate-direction leading-order ratio < 1 at operating intensity; the full worst-direction operator norm, a rigorous p_4 bound, off-anchor robustness, and RES-5 remain. The BCC $R=0.916$ is outside the operating regime ($A=0.08 \gg A_{\text{op}}=0.013$).
Next required action:	Bound the full worst-direction Hessian norm (the entropy cross-block) and the higher additive energies (p_4); then the lattice-class H-diag discharge (mirroring H-ADM-COH).